

Appendix A — Troubleshooting

Common build and runtime errors encountered when working with Zephyr 4.x on the nRF52840 DK, with their causes and fixes.

Build errors

Could NOT find Zephyr-sdk compatible with version 1.0

Cause: SDK 0.16.x or 0.17.x is installed instead of SDK 1.0.1. **Fix:**

```
ls ~/zephyr-sdk-* # check installed versions
cd ~
wget https://github.com/zephyrproject-rtos/sdk-ng/releases/
download/v1.0.1/zephyr-sdk-1.0.1_linux-x86_64_gnu.tar.xz
tar xf zephyr-sdk-1.0.1_linux-x86_64_gnu.tar.xz
cd zephyr-sdk-1.0.1 && ./setup.sh
```

Could NOT find Python3 (found version "3.10")

Cause: Zephyr 4.x requires Python 3.12. **Fix:**

```
python3.12 -m venv .venv
source .venv/bin/activate
pip install west
```

west: command not found

Cause: Virtual environment not activated. **Fix:**

```
source /home/jovin/workspace/My-Zephyr-project/myworkspace/.venv/
bin/activate
```

ZEPHYR_BASE is not set

Cause: ZEPHYR_BASE environment variable missing from the current shell. **Fix:**

```
export ZEPHYR_BASE=/home/jovin/workspace/My-Zephyr-project/  
myworkspace/zephyr
```

DT_NODELABEL(...) node does not exist

Cause: The device tree label does not match the board's .dts file. **Fix:** Check zephyr/boards/nordic/nrf52840dk/nrf52840dk_nrf52840.dts for the correct node label. Use a .overlay file to add or override nodes in your app.

undefined reference to 'bt_enable'

Cause: CONFIG_BT=y not set in prj.conf. **Fix:** Add the required Kconfig for the feature being used. BLE minimum: CONFIG_BT=y plus CONFIG_BT_PERIPHERAL=y or CONFIG_BT_CENTRAL=y.

***** BUS FAULT *** Imprecise data bus error on every boot**

Cause: Two issues that affect all projects on Zephyr 4.x:

1. CONFIG_COMMON_LIBC_MALLOC_ARENA_SIZE=-1 (the default) tells Zephyr to use all remaining RAM for the libc malloc heap. On Zephyr 4.4.x the “remaining RAM” calculation overshoots the 256 KB boundary by a few bytes, causing sys_heap_init to write its tail chunk past 0x20040000 — a bus fault.
2. void main(void) is invalid in Zephyr 4.x — main() must return int. Using void can corrupt the stack on return.

Fix: Add to every prj.conf:

```
CONFIG_COMMON_LIBC_MALLOC_ARENA_SIZE=0
```

And change every main.c signature:

```
/* wrong – Zephyr 4.x requires int */  
void main(void) { ... }  
  
/* correct */  
int main(void) { ...; return 0; }
```

Also remove CONFIG_STDOUT_CONSOLE=y if present — it enables a printf path that allocates from the same heap and is not needed when using printk.

Diagnostic: Open the .map file and check that sys_heap_init is at the faulting PC address. Confirm with:

```
grep "sys_heap_init" build/zephyr/zephyr.map
```

Runtime and flashing errors

LIBUSB_ERROR_ACCESS when running west flash

Cause: User lacks permissions to access the J-Link USB device. **Fix:**

```
sudo usermod -aG plugdev $USER
newgrp plugdev # apply without logging out
```

Serial terminal shows garbage or no output

Cause: Wrong baud rate (Zephyr default is 115200) or wrong port. **Fix:**

```
ls /dev/ttyACM*
screen /dev/ttyACM0 115200
# or
minicom -D /dev/ttyACM0 -b 115200
```

Application crashes with * USAGE FAULT *****

Cause: Stack overflow, null pointer dereference, or misaligned access.

Fix: Enable stack sentinel in prj.conf:

```
CONFIG_STACK_SENTINEL=y
CONFIG_THREAD_ANALYZER=y
```

Check logs for “Stack sentinel violated” or per-thread stack usage output.

sensor_sample_fetch returns -5 (EIO)

Cause: I2C address mismatch or SDA/SCL wiring error. **Fix:** 1. Verify the sensor’s I2C address in the overlay (e.g., reg = <0x0e> for KXTJ3). 2. Check wiring: SDA → P0.26, SCL → P0.27 on the nRF52840 DK. 3. Use a logic analyser or the Zephyr shell `i2c scan i2c@40003000` to detect the sensor.

BLE not advertising — no device in nRF Connect

Cause: `bt_enable` or `bt_le_adv_start` returned an error, or `CONFIG_BT_PERIPHERAL=y` is missing. **Fix:** Add logging and check for error codes:

```
int ret = bt_enable(NULL);  
if (ret < 0) { LOG_ERR("bt_enable: %d", ret); }
```

-22 (EINVAL) means invalid advertising parameters.

NVS write returns -28 (ENOSPC)

Cause: NVS partition is full or `sector_count` is too small. **Fix:** Increase `sector_count` in the `nvs_fs` struct, or call `nvs_clear(&fs)` to erase all entries and start fresh.

Quick reference: common errno values

Code	Value	Meaning
ENODEV	-19	Device not found or not ready
EINVAL	-22	Invalid argument
ENOSPC	-28	No space left (NVS full)
EIO	-5	I/O error (sensor comms failure)
EBUSY	-16	Device busy
ENOMEM	-12	Out of memory
ETIMEDOUT	-110	Operation timed out